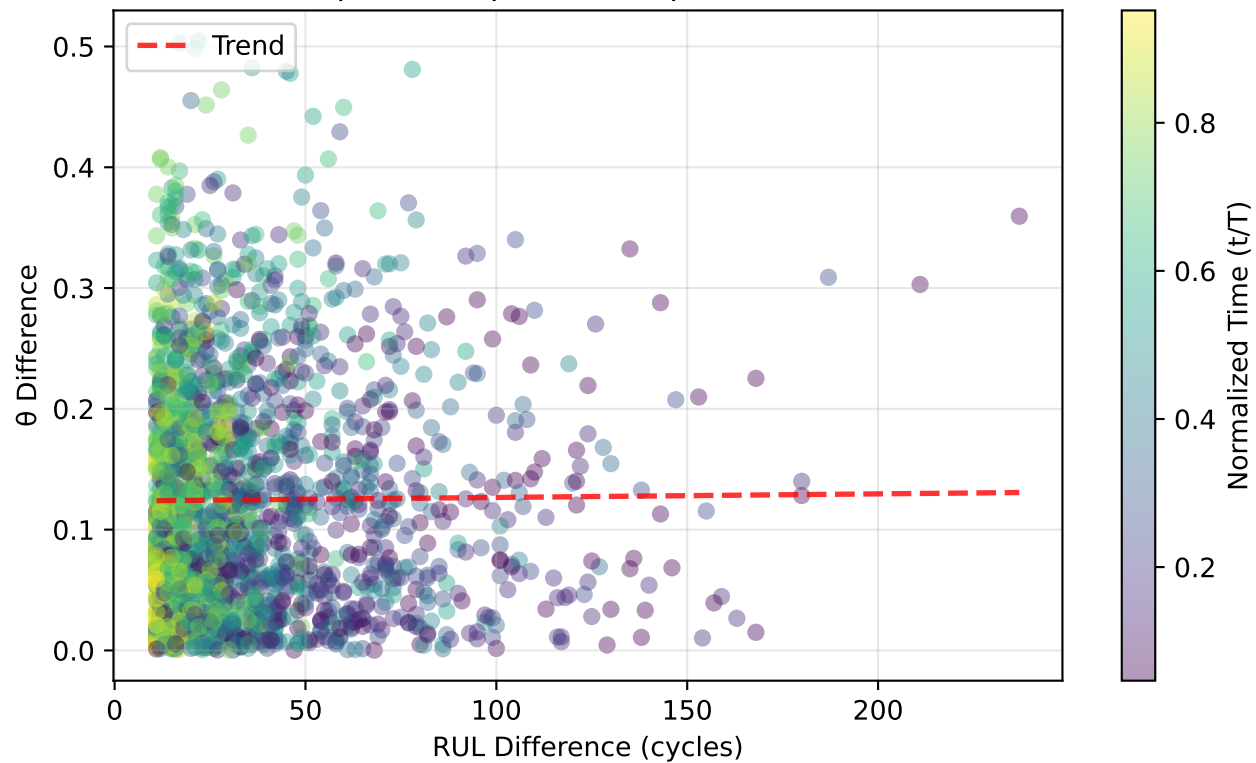
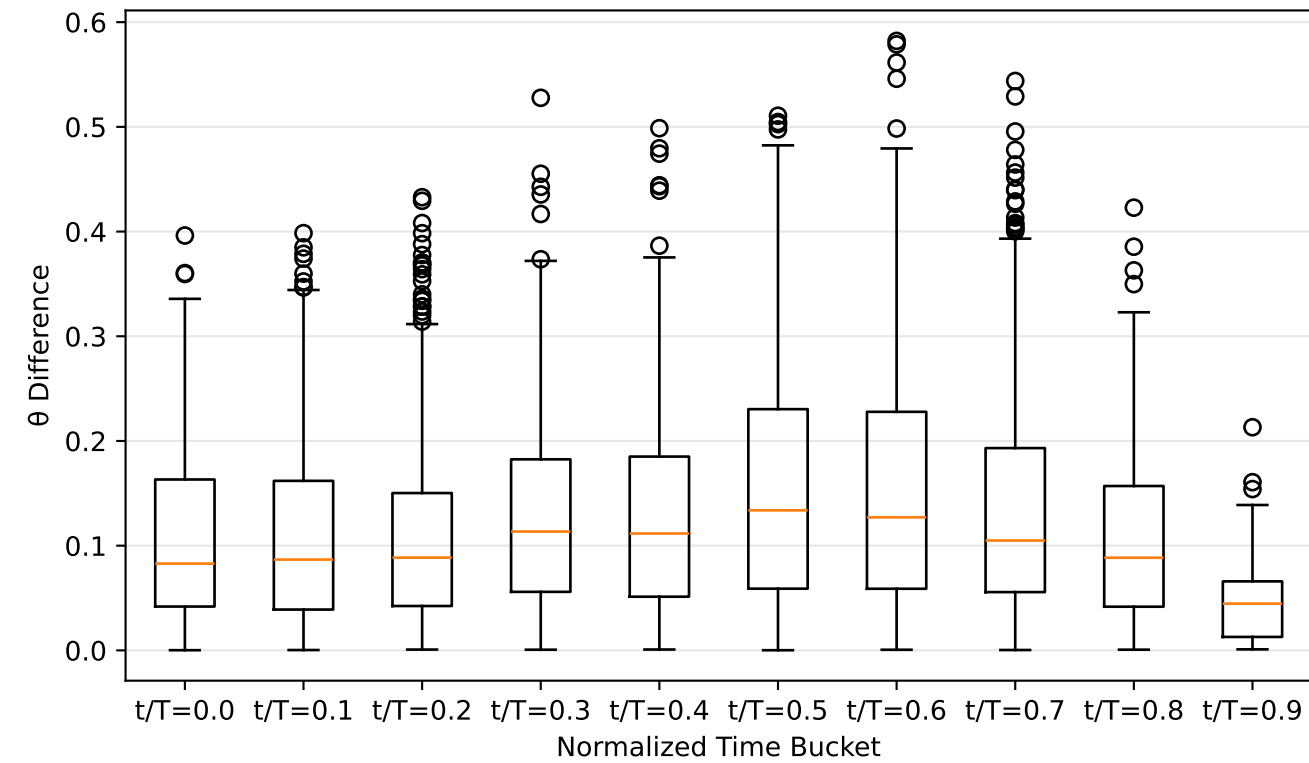


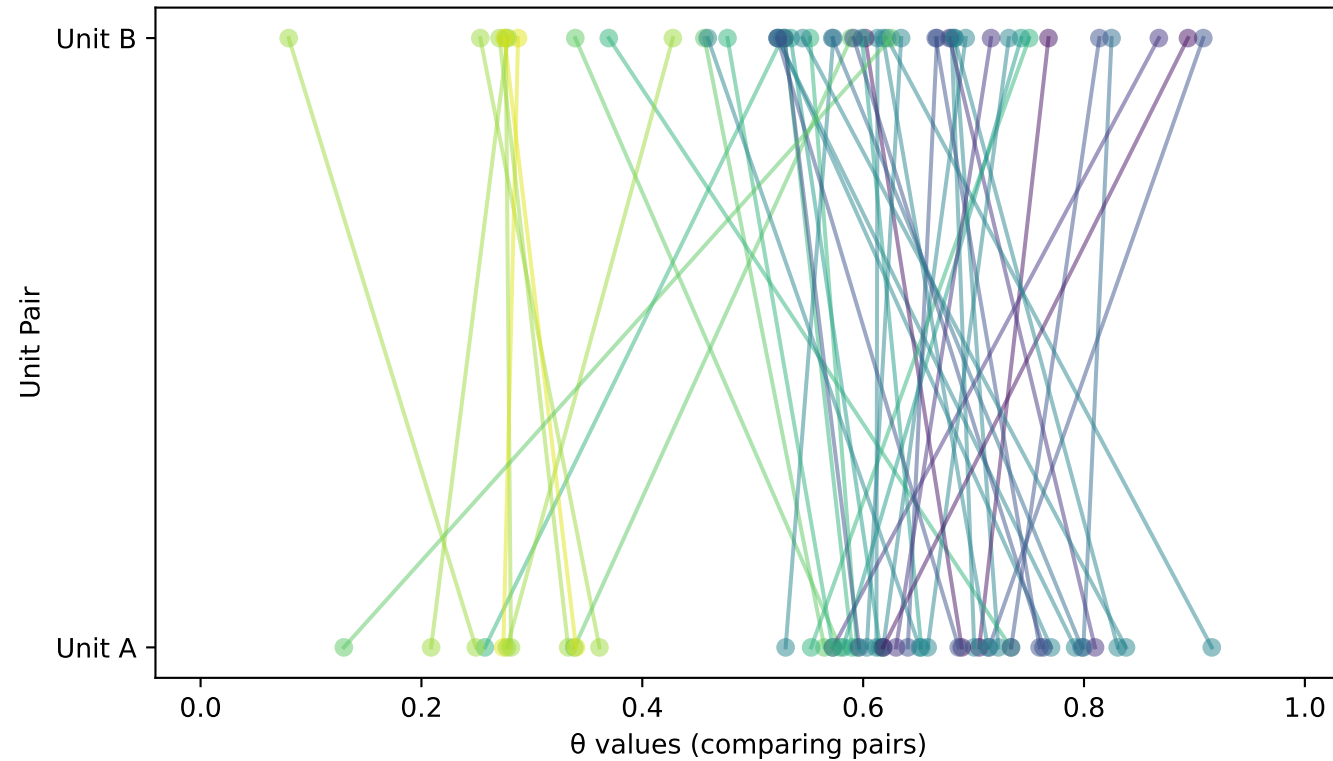
Health State Difference vs θ Difference
(Spearman $\rho=0.0374$, $p=1.11e-02$)



θ Difference Across Different Times



Sample Pairs: Showing θ Differences Within Pairs



Test 1c: Same-Time Different-Health Analysis

Dataset: FD002
Latent Dim: 4
Window Size: 10

Results:

Comparison Pairs: 4596

RUL Difference:
Mean: 35.4 cycles
Median: 26.0 cycles
Max: 237.0 cycles

θ Difference:
Mean: 0.1247
Median: 0.1005
Max: 0.5821

Correlation:
Spearman $\rho = 0.0374$
 p -value = $1.11e-02$

Interpretation:

If $\rho \gg 0$ and $p < 0.05$:
→ θ TRACKS STATE (different health leads to different θ even at same t/T)

If $\rho \approx 0$ or $p > 0.05$:
→ θ may be learning just t/T (health difference doesn't affect θ)